A register is a group of flip-flops used to store binary information temporarily. An n-bit register consists of n flip-flops and can store n bits of data.

Registers that are capable of moving information position wise upon the occurrence of a clock signal are called shift registers. A shift register is a sequential circuit that stores binary data and shifts it from one flip-flop to another for every clock pulse.

These registers are normally classified by whether they can move the information in one or two directions, i.e Unidirectional or Bidirectional.

The manner in which information is entered into and outputted from a register is another way in which they are categorized [SISO,SIPO,PIPO,PIPO]

Major applications of Registers…

1. Temporary Data Storage: Stores intermediate data during arithmetic and logical operations in a CPU.
2. Data Transfer: Transfers binary data between different sections of a processor.
3. Instruction Storage: Instruction Register (IR) stores the currently executing instruction.
4. Address Storage: Memory Address Register (MAR) holds the address of the memory location to be accessed.
5. Accumulator: Stores intermediate results of arithmetic and logical operations.
6. Buffering: Temporarily holds data during transfer between devices operating at different speeds.
7. Processor Operations: General-purpose registers store operands, addresses and intermediate results.

Major applications of Shift Registers…

1. Serial-to-Parallel Conversion: Converts serial data received from communication channels into parallel data using SIPO registers.
2. Parallel-to-Serial Conversion: Converts parallel data into serial form for transmission using PISO registers.
3. Temporary Data Storage: Stores binary information temporarily during digital operations.
4. Time Delay Circuits: Provides a digital delay by shifting data through successive flip-flops.
5. Sequence Generation: Generates predefined binary sequences using ring counters and Johnson counters.
6. Arithmetic Operations: Performs serial addition, multiplication and division by shifting binary data.
7. Data Transfer: Transfers data between digital systems and communication interfaces.
8. Digital Signal Processing: Used in digital filters, delay lines and signal-processing circuits.
9. LED Display Control: Controls running LED patterns and sequential lighting applications.
10. Pseudo-Random Sequence Generation: Linear Feedback Shift Registers (LFSRs) generate pseudo-random binary sequences used in testing and communication systems.

Based on the direction of data transfer and the method of data input/output, **shift registers operate in four basic modes.**

1. Serial-In Serial-Out (SISO) : Data input: Serial, Data output: Serial

- Data is entered into the register one bit at a time through a single input line.
- Each clock pulse shifts the data by one position.
- Data is obtained one bit at a time from the output of the last flip-flop.
- It is mainly used for **serial data transfer and time delay applications**.

Used as Circular SR, by connecting serial data out line to serial data in line

2. Serial-In Parallel-Out (SIPO) : Data input: Serial, Data output: Parallel

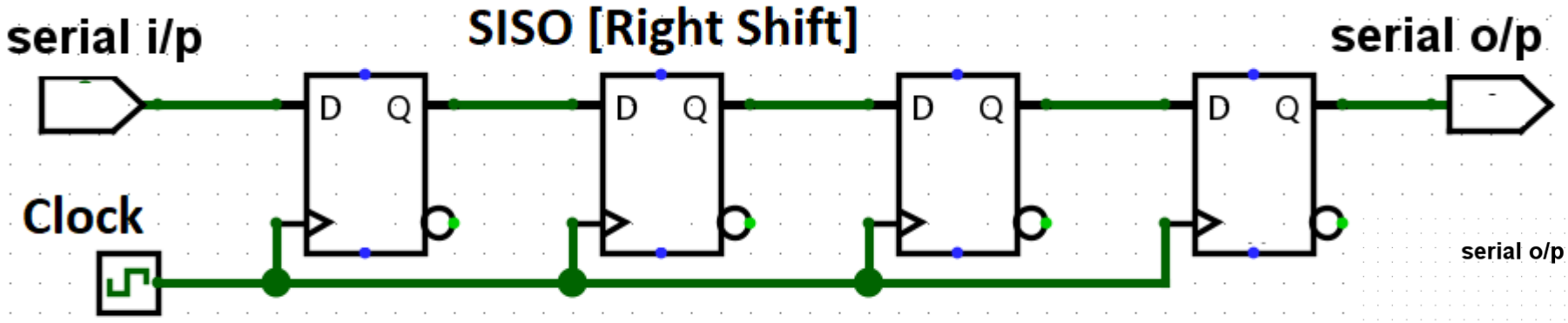
- Data is entered serially, one bit at a time.
- Each clock pulse shifts the incoming bit into the register.
- After all bits are loaded, the entire data is available simultaneously at the outputs of all flip-flops.
- It is used **for serial-to-parallel data conversion**.

3. Parallel-In Serial-Out (PISO): Data input: Parallel, Data output: Serial

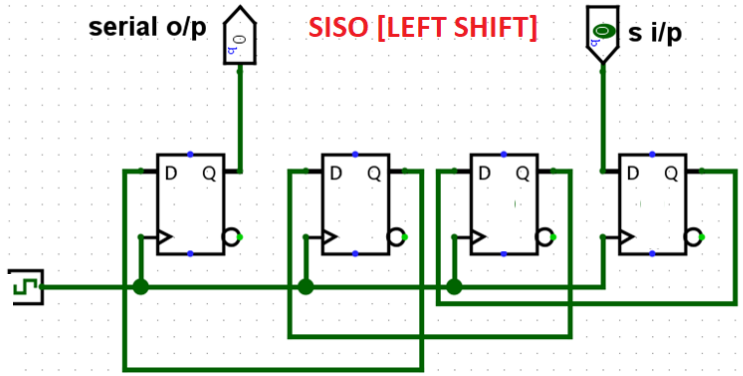
- All data bits are loaded simultaneously into the register using parallel inputs.
- A control signal selects parallel loading or shifting.
- After loading, each clock pulse shifts one bit towards the serial output.
- It is used for **parallel-to-serial data conversion.**

4. Parallel-In Parallel-Out (PIPO) : Data input: Parallel, Data output: Parallel

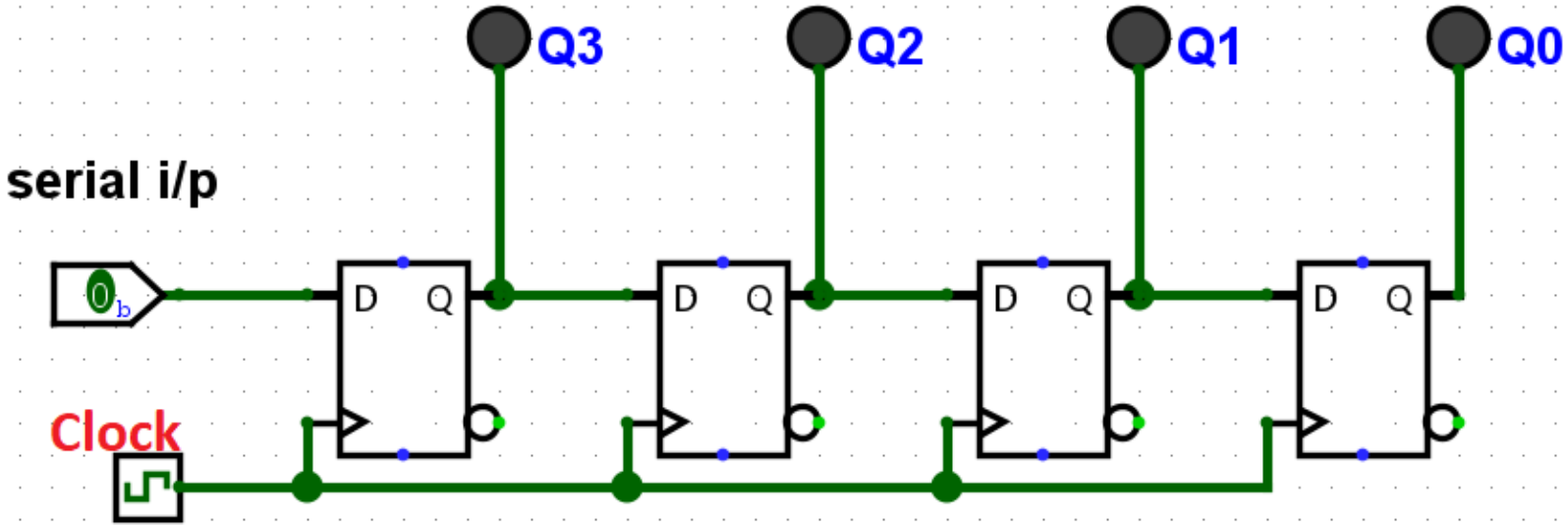
- All bits are loaded simultaneously into the register.
- All bits are available simultaneously at the output.
- No shifting of data is required during normal operation.
- It is mainly used for **temporary data storage and transfer**.

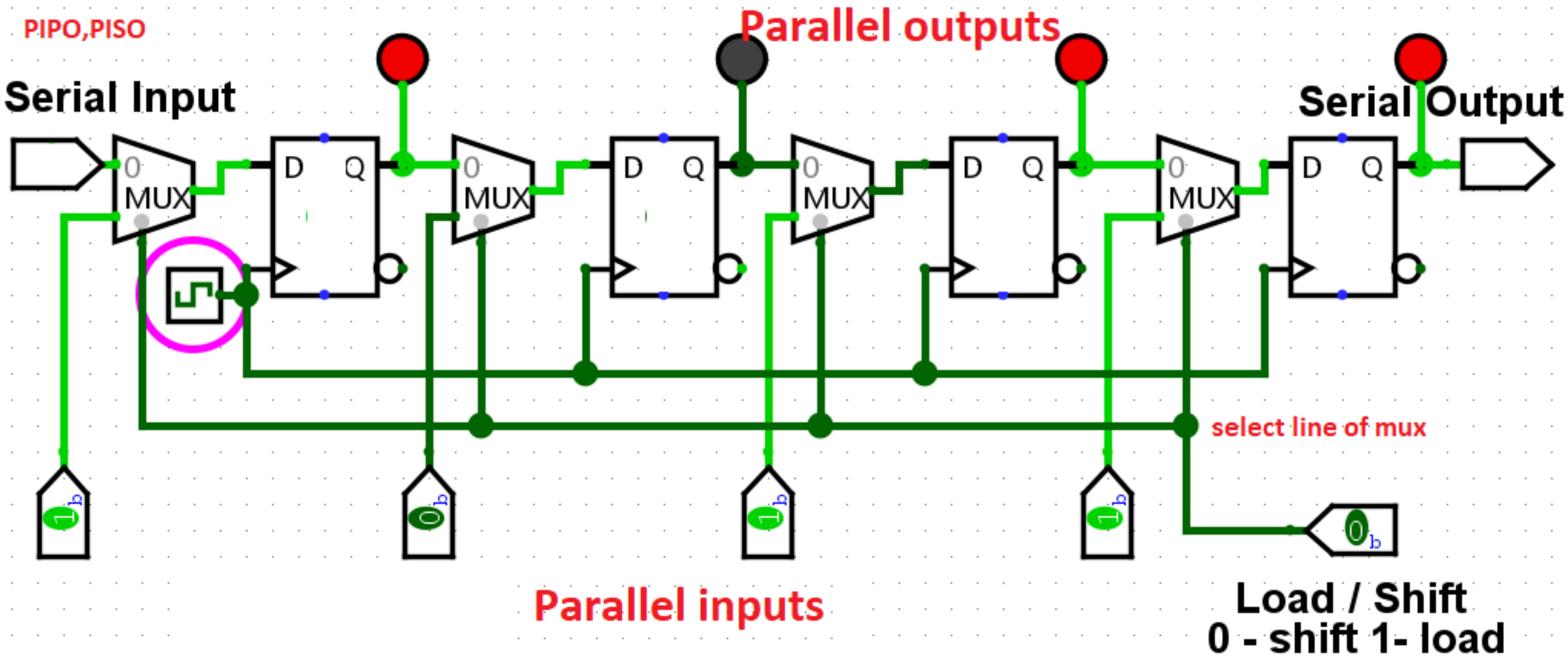


Note: Connect serial output to serial input, to realize the circular shift register



SIPO





Select Line of Multiplexer acts as Mode control pin:

0 – Means, it works in Shift mode. Data shifted out (right) one bit at a time, for every clock pulse

1 - Means, it works in Load mode. Parallel data is loaded into all the FlipFlops at a time, when clock pulse is applied.

Comparison of Four Modes

Mode	Input	Output	Main application
SISO	Serial	Serial	Data delay
SIPO	Serial	Parallel	Serial-to-parallel conversion
PISO	Parallel	Serial	Parallel-to-serial conversion
PIPO	Parallel	Parallel	Data storage

A circular shift register is a shift register in which the output of the last flip-flop is connected back to the input of the first flip-flop, forming a closed loop. As a result, the data stored in the register continuously circulates from one flip-flop to the next with every clock pulse.

In a 4-bit circular shift register:

- Four D flip-flops are connected in cascade.
- The output of the last flip-flop is fed back to the input of the first flip-flop.
- All flip-flops receive a common clock signal.

Universal Shift Register

A Universal Shift Register (USR) is a sequential logic circuit that can perform bidirectional shifting and supports SISO,SIPO,PISO,PIPO.

1. A 4-bit universal shift register consists of:

- Four D flip-flops: FF3, FF2, FF1 and FF0 (whose o/ps are Q3,Q2,Q1 and Q0)
- Four 4:1 multiplexers.
- Two mode-control inputs: S1 and S0.
- Four parallel data inputs: P3, P2, P1 and P0.
- Serial inputs for shifting in both directions.
- A common clock signal.

2. Multiplexer connections

Each D flip-flop is preceded by a 4:1 MUX. The MUX selects one of four possible inputs based on the mode-control signals S1 and S0.

S1	S0	Selected input	Operation
0	0	Present output Q	Hold
0	1	Adjacent higher bit	Shift right
1	0	Adjacent lower bit	Shift left
1	1	Parallel input P	Parallel load

3.Operation of the Universal Shift Register

Assume the register contains four bits: Q3,Q2,Q1,Q0

A. Hold operation (S1S0 = 00)

The output of each flip-flop is fed back to its own input through the MUX.
On the active clock edge, the existing data remains unchanged.

$$Q_3^+ Q_2^+ Q_1^+ Q_0^+ = Q_3 Q_2 Q_1 Q_0$$

B. Shift-right operation (S1S0 = 01)

Each bit shifts one position towards the right (LSB).
The serial input SR enters the MSB position.
The LSB is shifted out.

$$Q_3^+ = SR$$
$$Q_2^+ = Q_3$$
$$Q_1^+ = Q_2$$
$$Q_0^+ = Q_1$$

C. Shift-left operation (S1S0 = 10)

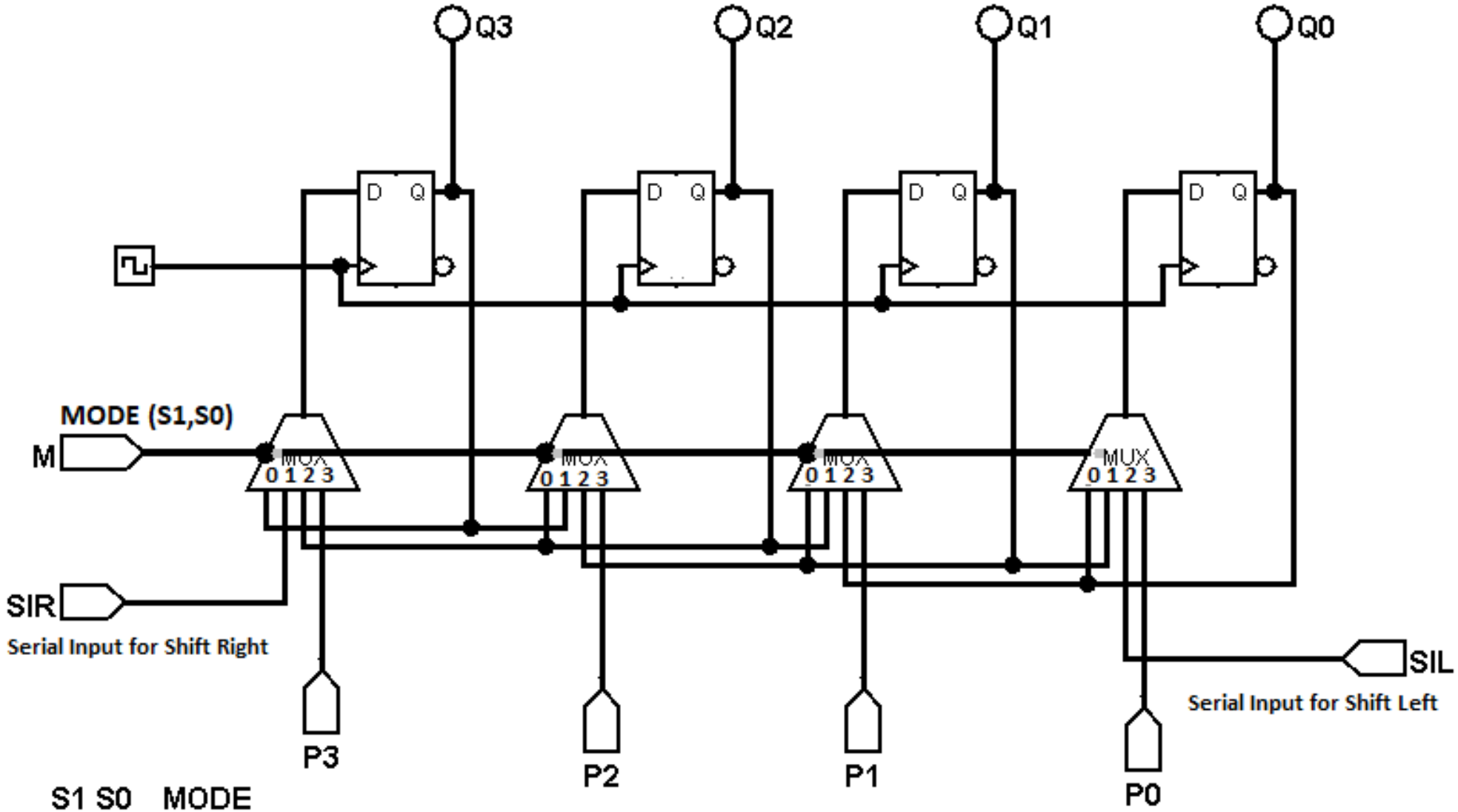
Each bit shifts one position towards the left (MSB).
The serial input SL enters the LSB position.
The MSB is shifted out.

$$Q_3^+ = Q_2$$
$$Q_2^+ = Q_1$$
$$Q_1^+ = Q_0$$
$$Q_0^+ \downarrow SL$$

D. Parallel-load operation (S1S0 = 11)

All four bits are loaded simultaneously.
Each flip-flop receives its corresponding parallel input.

$$Q_3^+ = P_3$$
$$Q_2^+ = P_2$$
$$Q_1^+ = P_1$$
$$Q_0^+ = P_0$$



S1	S0	MODE
0	0	HOLD
0	1	RIGHT SHIFT
1	0	LEFT SHIFT
1	1	PARALLEL LOAD

4 bit Universal Shift Register

Operation	Control	Register content
Initial state	–	1011
Hold	00	1011
Shift right (SR = 0)	01	0101
Shift left (SL = 0)	10	0110
Parallel load (P = 1100)	11	1100

CIE2, Year 2025

Construct the logic diagram of a 4-bit register with four D flip-flops and 4 × 1 multiplexers with mode selection input m1 and m0. The register operates according to the following function table:

m1	m0	Register operation
0	0	No Change
0	1	Complement the four Output
1	0	Clear register to 0 (Synch)
1	1	Load parallel data

Ans:

Hint: Connect Q* (complimented Q) to D inputs, for Mode (01) ; Connect 0 to Input 2 of the Multiplexers for Synchronous Clear operation.

CIE2, Year 2024

Design a 4 bit Universal Shift Register, Which supports following modes of operation-

S1	S0	Operations
0	0	Hold
0	1	Shift Right Data
1	0	Circular Shift Left Data
1	1	Parallel Load the Data

Ans:

Hint: Q3 to be connected SIL

Only for Reference: **Universal Shift Register – Circuit given as per Givone book**

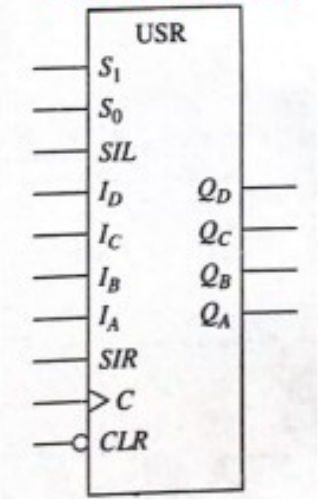
A Universal Shift Register (USR) is a sequential logic circuit that can perform bidirectional shifting and supports SISO,SIPO,PISO,PIPO. The following USR design four basic register operations:

1.Hold (no change in data).
2.Shift right.
3.Shift left.
4.Parallel load.

Modes of Operation

Select lines S_1	S_0	Register operation
0	0	Hold
0	1	Shift right
1	0	Shift left
1	1	Parallel load

Logic Symbol



4bit Universal Shift Register

